AEOS GUARD

Agentic Execution and Operations Security

A Unified Security Middleware for Autonomous AI Systems

Project Overview · For academic review

Prepared for classroom presentation

What is AEOS Guard?

AEOS Guard is a **security layer that sits around an AI agent** and watches everything it does, in real time, to catch attacks before they cause harm.

Modern AI agents don't just answer questions — they take actions on their own. They use tools, look things up in a memory, and reason through problems step by step. That freedom is what makes them useful, but it also gives attackers three separate doors to sneak through. AEOS Guard is a security guard standing at all three doors at once, and it reports everything it sees to a single live dashboard.

In one sentence

> *"AEOS Guard protects an AI agent's **tools**, its **memory**, and its **reasoning** — the three places it can be attacked — and shows a security team exactly what was attempted and what was stopped."*

Who is it for?

- **Companies building AI agents** that act on their own — customer-service bots, coding assistants, research agents — and need to trust what those agents do.
- **Security teams** who currently have no single place to see what is being attempted against their AI systems.
- **Anyone deploying an AI agent** in the real world, where the tools and data it relies on cannot all be assumed safe.

The Problem

An autonomous AI agent can be attacked in more than one place — and protecting only one place leaves the others wide open. Today, most defences look at a single layer. AEOS Guard is built on the idea that a real attacker won't politely stick to one.

TOOL LAYER

The tools get swapped

An agent approves a tool that looks safe. Later, the tool quietly changes its own instructions and turns malicious — a “rug pull.” The agent never notices.

MEMORY LAYER

The memory gets poisoned

An attacker plants a booby-trapped document in the agent's memory, cleverly written to get pulled into almost every conversation — no matter what is being asked.

REASONING LAYER

The thinking gets hijacked

The agent can be steered into unsafe reasoning, or coaxed into leaking a password or private detail in the middle of its own thought process.

Why this is hard

- The three attacks look completely unrelated — they happen in different places, at different times, in different formats.
- A tool that hardens only one door gives a false sense of safety.
- Even when each attack is spotted, nobody can see that they are **the same attacker**, running one coordinated campaign against one agent.

The core problem: three separate blind spots, and no single view that connects them.

The Solution

AEOS Guard places an independent **defender at each of the three doors**, and connects them to **one dashboard** that shows the whole picture. Each defender works on its own, but they all speak the same language — so the dashboard can tell when the same agent is being hit at more than one layer.

MCP-Shield TOOL LAYER

Takes a secure “fingerprint” of every tool when it is first approved. If a tool later changes, MCP-Shield spots the mismatch and **blocks the tampered version**, handing the agent back the original, trusted one.

VectorAnchor MEMORY LAYER

Watches which documents keep getting pulled into unrelated conversations. A normal document is relevant to one topic; a booby-trap shows up everywhere — so VectorAnchor **quarantines it** and serves a clean result instead.

TraceAudit REASONING LAYER

Reads the agent's reasoning as it streams out. If the thinking drifts somewhere unsafe it **stops the response**; if a password or private detail appears, it **blacks it out** before anyone can see it.

The unifying idea

Every defender reports what it sees in the **same simple format** to one live dashboard. That shared language is what turns three separate tools into a single system — and it is what lets AEOS Guard say the thing no single defender can: **“these three alerts are the same attacker, hitting the same agent, in one session.”**

How It Works

Every defender follows the same four-step rhythm, no matter which door it is guarding:

1

WATCH

Sit directly on the path the agent uses — nothing gets past unseen.

2

CHECK

Compare what's happening against what is known to be safe.

3

ACT

Block, quarantine, or black out anything dangerous — automatically.

4

REPORT

Send a clear alert to the shared dashboard, tagged to the agent.

The dashboard ties it together

All three defenders feed one real-time screen. A security team sees every detection as it happens, colour-coded by layer and by how serious it is. They can then take ownership of an alert, investigate it, and mark it as handled — with a permanent, tamper-proof record of who did what and why.

The headline feature

If the tool defender, the memory defender, and the reasoning defender all raise an alert for the **same agent session**, the dashboard flags it as **one coordinated attack on one compromised agent** — not three unrelated blips. That connected view is the whole point of the project.

Problems Solved, at a Glance

Layer	The attack	What AEOS Guard does
Tool	A trusted tool is secretly swapped for a malicious one.	Fingerprints every tool and blocks any tampered version.
Memory	A poisoned document is planted to reach every conversation.	Spots the over-appearing document and quarantines it.
Reasoning	The agent is steered off-course or leaks a secret.	Stops unsafe reasoning and blacks out leaked secrets.
All three	The same attacker hits several layers and looks unrelated.	Links the alerts to one agent session and one campaign.

What makes it stand out

- **Defence in depth.** Three independent guards, not one — breaking through one does not get you past the others.
- **One connected picture.** The first system that can say three separate alerts are the same attacker.
- **Never loses an alert.** If the dashboard is briefly down, every detection is saved and delivered when it comes back — a security tool must not drop evidence.
- **Runs on your own machine.** No cloud service required; the whole system starts with a single command and works fully offline.
- **Honest about its limits.** Every claim is backed by a test, and the known ways to slip past it are measured and written down — not hidden.

Where the Project Stands

AEOS Guard is a working system, not a slide deck. All three defenders run, the dashboard is live, and the whole thing can be started with one command and driven through a full demonstration attack from end to end.

3 layers
guarded independently — tool, memory, reasoning

1 dashboard
one live, connected view of every attack

100+
automated checks proving the defences work

Proven end to end

- A full demonstration launches all three attacks and confirms every one is caught, recorded, and traced back to a single compromised agent session.
- The defenders add almost no delay — the checks take microseconds, far too fast for a user to notice.
- The known ways an attacker could still slip past are deliberately tested and measured, so the project is honest about exactly how far its protection reaches.

In short

AEOS Guard makes an autonomous AI agent safe to trust — by guarding all three places it can be attacked, catching the attacks automatically, and giving a security team one clear, connected view of everything that was attempted and everything that was stopped.